

REPRODUCTION IN PLANTS

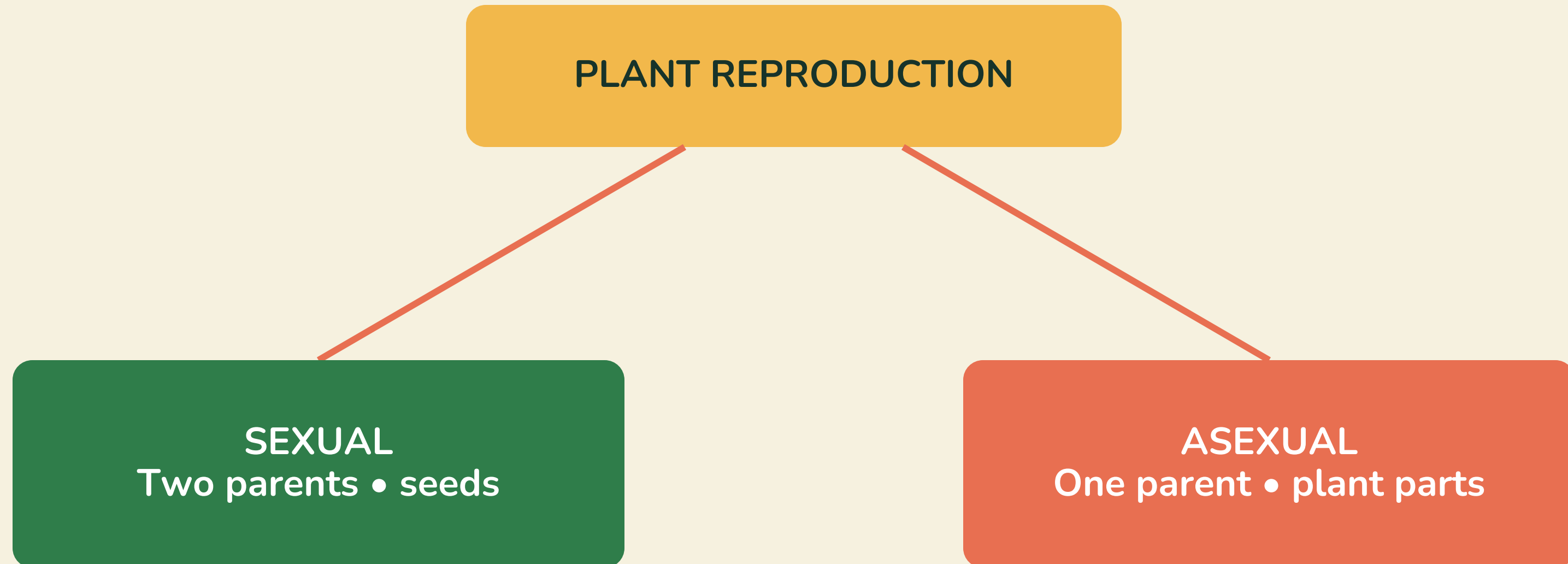
How plants grow new life

Look closely. Every new plant has a story.



What is reproduction?

Reproduction is how living things make new individuals of their kind.



Pollination: pollen on the move

Pollination is the transfer of pollen from male to female parts of a flower.

MALE PART

Stamen makes pollen

FEMALE PART

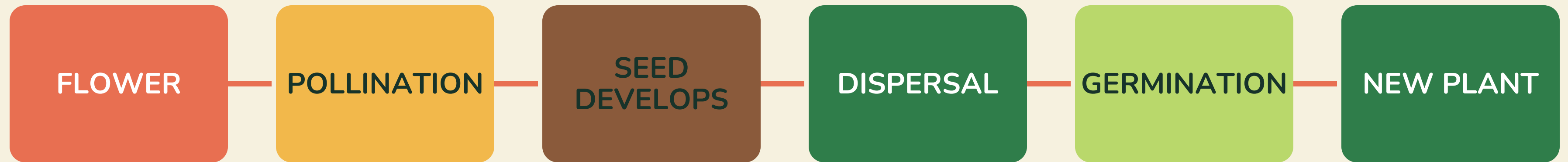
Stigma receives pollen

Agents: insects • birds • wind •
other animals



From flower to new plant

Follow the journey. What must happen before a seed can sprout?



A seed needs: WATER AIR WARMTH

One parent. Four clever methods.

New plants grow from parts of a parent plant — without seeds.

CUTTING
Stem or leaf

BUDDING
A bud joins

LAYERING
Stem roots

GRAFTING
Two parts join

Which method have you seen in a garden?

Cutting

Take a stem or leaf piece. Place it in soil or water. It grows roots and becomes a new plant.

1 Choose a healthy stem

2 Place it in soil

3 Roots develop

LOOK FOR: a node — the spot where a leaf joins the stem.



Budding

A bud from one plant is attached to another plant. Growing tissue inside the bud forms a new shoot.

BUD PIECE

HOST STEM



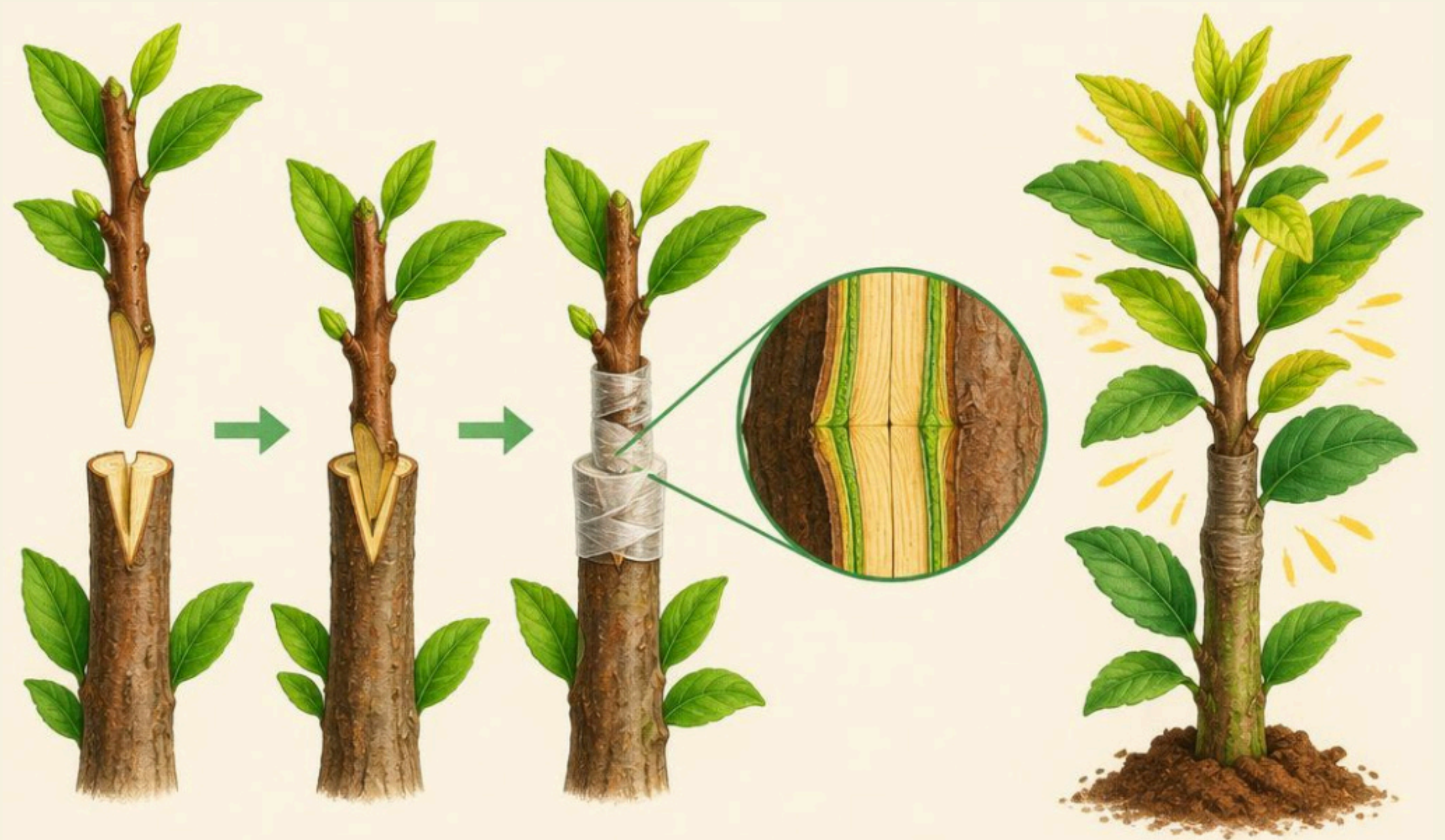
Grafting

Parts of two plants are joined so they grow together as one — often to improve a plant variety.

SCION
The upper shoot

ROOTSTOCK
The rooted base

GRAFT UNION
Where the parts heal together



Compare the methods

Notice what starts each new plant.

METHOD	WHAT HAPPENS	PRODUCES
Pollination	Pollen reaches stigma	Seeds
Cutting	Stem/leaf grows roots	Clone
Budding	Bud joins host	Joined plant
Layering	Attached stem roots	Clone
Grafting	Two plant parts join	Joined plant

WHY IT MATTERS

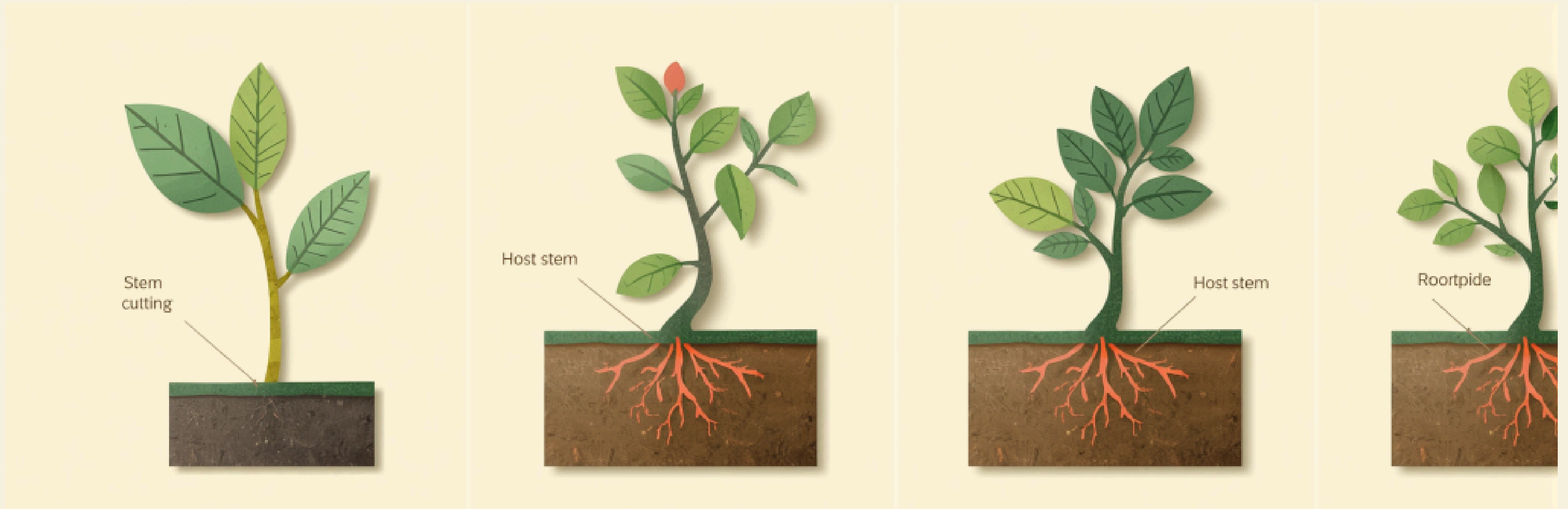
New life keeps nature moving.

- Produces new plants
- Continues the species
- Increases populations
- Spreads plants to new places



Activity 1: What method is it?

Study each picture. Write A, B, C or D beside the correct method.



Can you name the method shown?

Activity 2: Arrange the process

These cards are mixed up. Point to them in the correct order.

GERMINATION

POLLINATION

NEW PLANT

SEED DISPERSAL

SEED DEVELOPMENT

Hint: What must happen first?

Activity 3: Seed or plant part?

Move each method in your mind to the correct side.

BY SEED

BY PLANT PART

POLLINATION + SEED

CUTTING

BUDDING

LAYERING

GRAFTING

Check your understanding

Choose the best answer for each question.

1 What does pollination transfer?
A.roots B.pollen C.water

2 Which method uses a stem piece?
A.cutting B.wind C.seed

3 In layering, when do roots grow?
A.while attached B.after flowering C.never

4 What joins in grafting?
A.two plant parts B.two seeds
C.two flowers

5 What does germination begin?
A.a new plant B.pollination
C.grafting

Answer key

Check your thinking — and explain why each answer makes sense.

✓ 1 B — pollen is transferred

✓ 2 A — a cutting uses a stem or leaf

✓ 3 A — roots grow while attached

✓ 4 A — two plant parts are joined

✓ 5 A — germination begins a new plant

Five things to remember

Can you teach these ideas to a partner in one minute?

1

Reproduction makes new plants.

2

Pollination moves pollen to female parts.

3

Sexual reproduction produces seeds.

4

Asexual reproduction uses one parent's parts.

5

Reproduction helps plants survive and spread.

THANK YOU!

Any questions?

Reflect: Which reproduction method surprised you most?

